Additive and interaction models (answer key)

Each of Parts 1–3 has two scenarios with the same tasks. We'll work through scenario **A** together as a class. Then you'll work through scenario **B** in your group. In Part 4, you'll design studies of your own.

Every scenario asks the same question: does the effect of one predictor **depend on** the other? The **additive** model says no: each predictor's effect is the same at every value of the other. The **interaction** model adds one more coefficient that lets the effect of one predictor change with the other. Does that coefficient earn its df?

$$F = \frac{(\text{RSS}_{simple} - \text{RSS}_{complex}) \;/\; (p_{complex} - p_{simple})}{\text{RSS}_{complex} \;/\; (n - p_{complex})}$$

- RSS: residual sum of squares, $\sum(y_i - \hat{y}_i)^2$. The variation the model leaves unexplained.
- p: number of coefficients the model estimates, including the intercept
- n: number of observations
- $df = n - p$: degrees of freedom left over after fitting the model

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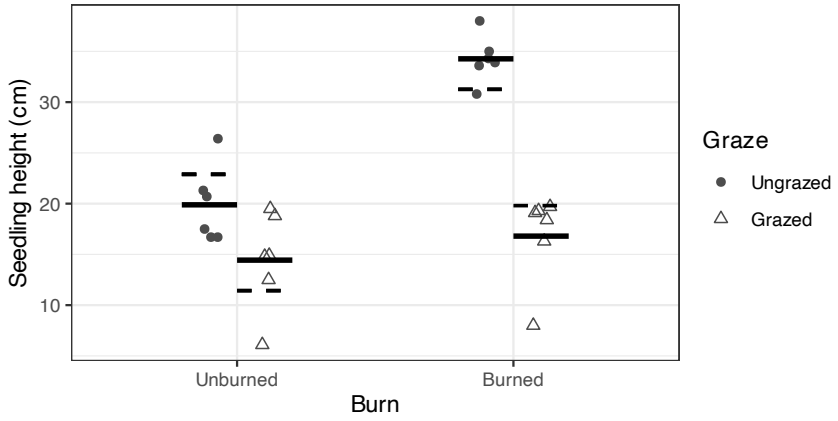
Part 1: Two categorical predictors

For each scenario:

1. **Data.** Look at the data in the figure.
2. **Full model.** Write the equation for the interaction model. What does the interaction coefficient mean?
3. **Draw the models.** Use the coefficients to draw each model's prediction for each combination as a horizontal segment: the interaction model (solid) and the additive model (dashed). How are they different?
4. **Degrees of freedom.** Fill in p and df for each model. How many df does the interaction cost?
5. **F test.** Find the F statistic and p-value for the interaction in the `anova()` table.
6. **Interpretation.** State H_0 and H_A. Then interpret the F test in a sentence about the system, not the coefficients.

1A. Fire and grazing (together)

Land managers use prescribed burns to clear dead vegetation and give native seedlings room to grow. But cattle also graze many of these lands. A range ecologist measured seedling **height** (cm) in 6 plots for each combination of **burn** (unburned or burned) and **graze** (ungrazed or grazed). Does the effect of burning depend on grazing?



Full model

$\text{height} = \beta_0 + \beta_1(\text{burn} = \text{Burned}) + \beta_2(\text{graze} = \text{Grazed}) + \beta_3(\text{burn} = \text{Burned})(\text{graze} = \text{Grazed}) + \varepsilon$

β_3 is how much the effect of burning changes in grazed plots, compared to ungrazed plots.

Draw the models

```
coef(lm(height ~ burn + graze, plots))
```

```
(Intercept)  burnBurned grazeGrazed
   22.88750     8.37500   -11.45833
```

```
coef(lm(height ~ burn * graze, plots))
```

```
          (Intercept)             burnBurned            grazeGrazed burnBurned:grazeGrazed
             19.88333              14.38333              -5.45000             -12.01667
```

Interaction model (solid): Unburned, Ungrazed = 19.9; Burned, Ungrazed = 34.3; Unburned, Grazed = 14.4; Burned, Grazed = 16.8 cm. These are the means of each combination. Additive model (dashed): Unburned, Ungrazed = 22.9; Burned, Ungrazed = 31.3; Unburned, Grazed = 11.4; Burned, Grazed = 19.8 cm. In the additive model, burning adds the same amount in grazed and ungrazed plots. The interaction model lets the effect of burning differ: burning makes a big difference in ungrazed plots and almost none in grazed plots.

Degrees of freedom (n = 24)

Model	p	df	RSS
`height ~ burn`	2	22	1,321.0
`height ~ burn + graze`	3	21	533.2
`height ~ burn * graze`	4	20	316.6

How many df does the interaction cost?

1 df: the interaction adds one coefficient, β_3.

F test

The F statistic and p-value for adding the interaction are in the `burn:graze` row.

```
anova(lm(height ~ burn * graze, plots))
```

```
Analysis of Variance Table

Response: height
           Df Sum Sq Mean Sq F value    Pr(>F)
burn        1 420.84  420.84  26.582 4.809e-05 ***
graze       1 787.76  787.76  49.758 7.693e-07 ***
burn:graze  1 216.60  216.60  13.681  0.001421 **
Residuals  20 316.64   15.83
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$ (the effect of burning is the same in grazed and ungrazed plots)

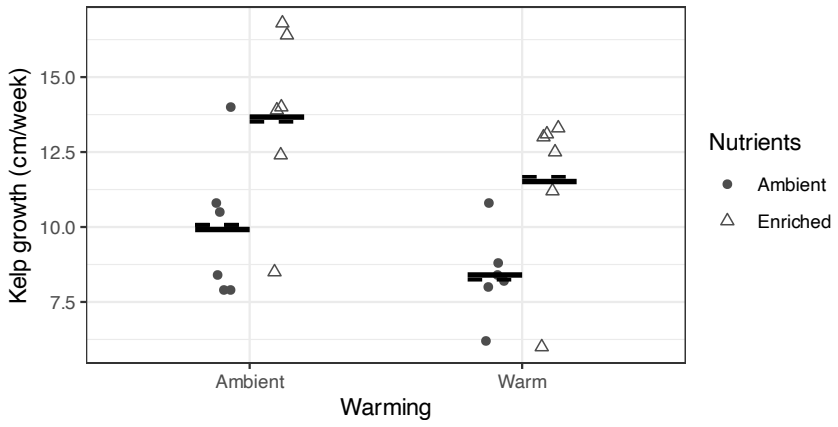
H_A: $\beta_3 \neq 0$

Interpret the F test for the interaction in a sentence.

If the effect of burning didn't depend on grazing, we'd see an F at least this large with probability p = 0.001. We reject H_0. Burning helps seedlings grow, but only where cattle don't graze.

1B. Warming and nutrients (in groups)

Marine heatwaves stress kelp forests, while nutrient-rich upwelling can boost kelp growth. Can nutrients offset warming? A lab grew kelp in 6 tanks for each combination of **warming** (ambient or warm water) and **nutrients** (ambient or enriched), and measured **growth** (cm/week). Does the effect of warming depend on nutrients?



Full model

growth $= \beta_0 + \beta_1(\text{warming} = \text{Warm}) + \beta_2(\text{nutrients} = \text{Enriched}) + \beta_3(\text{warming} = \text{Warm})(\text{nutrients} = \text{Enriched}) + \varepsilon$

β_3 is how much the effect of warming changes with enriched nutrients, compared to ambient nutrients.

Draw the models

```
coef(lm(growth ~ warming + nutrients, tanks))
```

```
      (Intercept)       warmingWarm nutrientsEnriched 
        10.075000         -1.833333          3.433333 
```

```
coef(lm(growth ~ warming * nutrients, tanks))
```

```
                 (Intercept)                   warmingWarm             nutrientsEnriched 
                   9.9166667                    -1.5166667                     3.7500000 
warmingWarm:nutrientsEnriched 
                  -0.6333333 
```

Interaction model (solid): Ambient, Ambient = 9.9; Warm, Ambient = 8.4; Ambient, Enriched = 13.7; Warm, Enriched = 11.5 cm/week. Additive model (dashed): Ambient, Ambient = 10.1; Warm, Ambient = 8.2; Ambient, Enriched = 13.5; Warm, Enriched = 11.7 cm/week. The two models' predictions are close: warming lowers growth by about the same amount at both nutrient levels.

Degrees of freedom (n = 24)

Model	p	df	RSS
`growth ~ warming`	2	22	195.7
`growth ~ warming + nutrients`	3	21	124.9
`growth ~ warming * nutrients`	4	20	124.3

How many df does the interaction cost?

1 df: the interaction adds one coefficient, β_3.

F test

The F statistic and p-value for adding the interaction are in the `warming:nutrients` row.

```
anova(lm(growth ~ warming * nutrients, tanks))
```

```
Analysis of Variance Table

Response: growth
                  Df  Sum Sq Mean Sq F value   Pr(>F)
warming            1  20.167  20.167  3.2441 0.086781 .
nutrients          1  70.727  70.727 11.3772 0.003024 **
warming:nutrients  1   0.602   0.602  0.0968 0.758941
Residuals         20 124.330   6.217
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$ (the effect of warming is the same at both nutrient levels)

H_A: $\beta_3 \neq 0$

Interpret the F test for the interaction in a sentence.

If the effect of warming didn't depend on nutrients, we'd see an F at least this large with probability p = 0.759. We fail to reject H_0. Warming slows kelp growth and nutrients speed it up, but nutrients don't change how much warming hurts. Nutrients offset warming by adding growth, not by protecting kelp from heat.

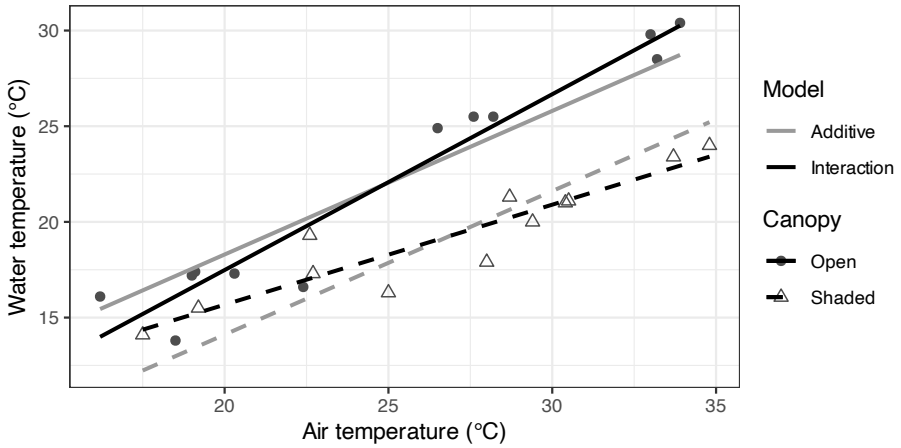
Part 2: One categorical and one continuous predictor

Now one predictor is continuous. In the additive model, the groups get parallel lines: the same slope, different intercepts. The interaction model gives each group its own slope. For each scenario:

1. **Data.** Look at the data in the figure.
2. **Full model.** Write the equation for the interaction model. What does the interaction coefficient mean?
3. **Draw the models.** Use the coefficients to draw the additive model (two parallel lines) and the interaction model (two lines with different slopes) on the figure.
4. **Degrees of freedom.** Fill in p and df for each model. How many df does the interaction cost?
5. **F test.** Find the F statistic and p-value for the interaction in the `anova()` table.
6. **Interpretation.** State H_0 and H_A. Then interpret the F test in a sentence about the system, not the coefficients.

2A. Stream temperature (together)

Trout need cold water. As summers get hotter, streams warm too, but trees along the banks shade the water. A watershed council measured summer **water** temperature (°C) and **air** temperature (°C) at 12 stream reaches with each type of **canopy** (open banks or shaded by trees). Does the effect of air temperature on water temperature depend on shade?



Full model

$$\text{water} = \beta_0 + \beta_1 \text{air} + \beta_2(\text{canopy} = \text{Shaded}) + \beta_3 \text{air} \times (\text{canopy} = \text{Shaded}) + \varepsilon$$

β_1 is the slope for open reaches. β_3 is how much the slope changes for shaded reaches: the shaded slope is $\beta_1 + \beta_3$.

Draw the models

```
coef(lm(water ~ air + canopy, streams))
```

```
 (Intercept)          air canopyShaded
   3.2904036    0.7503026   -4.1881204
```

```
coef(lm(water ~ air * canopy, streams))
```

```
     (Intercept)              air     canopyShaded air:canopyShaded
      -0.8911888        0.9187454        6.1124259       -0.3961248
```

Additive model (grey): Open: intercept 3.3, slope 0.75; Shaded: intercept -0.9, slope 0.75. Interaction model (black): Open: intercept -0.9, slope 0.92; Shaded: intercept 5.2, slope 0.52.

Degrees of freedom (n = 24)

Model	p	df	RSS
`water ~ air`	2	22	172.4
`water ~ air + canopy`	3	21	70.5
`water ~ air * canopy`	4	20	40.5

How many df does the interaction cost?

1 df: the interaction adds one coefficient, β_3.

F test

The F statistic and p-value for adding the interaction are in the `air:canopy` row.

```
anova(lm(water ~ air * canopy, streams))
```

```
Analysis of Variance Table

Response: water
           Df Sum Sq Mean Sq F value    Pr(>F)
air         1 379.64  379.64 187.342 1.286e-11 ***
canopy      1 101.95  101.95  50.309 7.101e-07 ***
air:canopy  1  29.94   29.94  14.774  0.001014 **
Residuals  20  40.53    2.03
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$ (water temperature rises with air temperature at the same rate in open and shaded reaches)

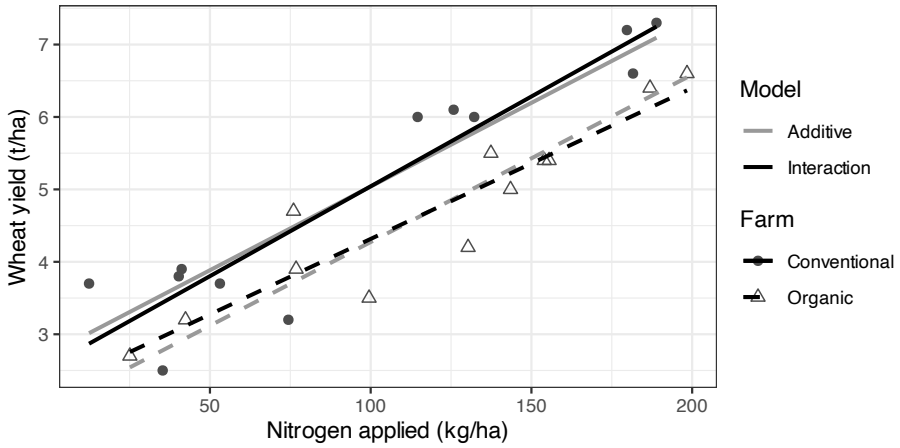
H_A: $\beta_3 \neq 0$

Interpret the F test for the interaction in a sentence.

If the effect of air temperature didn't depend on shade, we'd see an F at least this large with probability p = 0.001. We reject H_0. Open reaches warm by 0.92 °C for every 1 °C of air temperature, but shaded reaches warm by only 0.52 °C. Shade matters more the hotter the summer.

2B. Fertilizer and farm type (in groups)

Organic farms often yield less than conventional farms. Does nitrogen fertilizer close the gap? An agricultural extension office measured wheat **yield** (t/ha) and the **nitrogen** applied (kg/ha) on 12 fields of each **farm** type (conventional or organic). Does the effect of nitrogen on yield depend on farm type?



Full model

$\text{yield} = \beta_0 + \beta_1 \text{nitrogen} + \beta_2(\text{farm} = \text{Organic}) + \beta_3 \text{nitrogen} \times (\text{farm} = \text{Organic}) + \varepsilon$

β_1 is the slope for conventional farms. β_3 is how much the slope changes for organic farms: the organic slope is $\beta_1 + \beta_3$.

Draw the models

```
coef(lm(yield ~ nitrogen + farm, farms))
```

```
(Intercept)    nitrogen farmOrganic
 2.72839517  0.02311086 -0.76601704
```

```
coef(lm(yield ~ nitrogen * farm, farms))
```

```
        (Intercept)            nitrogen         farmOrganic nitrogen:farmOrganic
        2.561957357         0.024804164        -0.329988955        -0.003962266
```

Additive model (grey): Conventional: intercept 2.7, slope 0.023; Organic: intercept 2.0, slope 0.023. Interaction model (black): Conventional: intercept 2.6, slope 0.025; Organic: intercept 2.2, slope 0.021.

Degrees of freedom (n = 24)

Model	p	df	RSS
`yield ~ nitrogen`	2	22	10.1
`yield ~ nitrogen + farm`	3	21	6.7
`yield ~ nitrogen * farm`	4	20	6.4

How many df does the interaction cost?

1 df: the interaction adds one coefficient, β_3.

F test

The F statistic and p-value for adding the interaction are in the `nitrogen:farm` row.

```
anova(lm(yield ~ nitrogen * farm, farms))
```

```
Analysis of Variance Table

Response: yield
              Df Sum Sq Mean Sq  F value   Pr(>F)
nitrogen       1 38.635  38.635 121.1925 6.14e-10 ***
farm           1  3.410   3.410  10.6962 0.003827 **
nitrogen:farm  1  0.299   0.299   0.9372 0.344561
Residuals     20  6.376   0.319
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$ (yield rises with nitrogen at the same rate on conventional and organic farms)

H_A: $\beta_3 \neq 0$

Interpret the F test for the interaction in a sentence.

If the effect of nitrogen didn't depend on farm type, we'd see an F at least this large with probability p = 0.345. We fail to reject H_0. Nitrogen raises yield by about the same amount on both farm types, so adding nitrogen doesn't close the gap between organic and conventional farms.

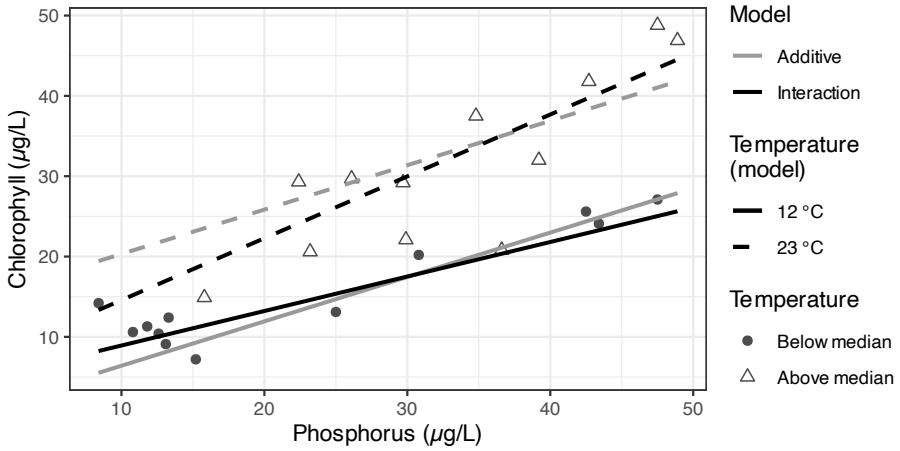
Part 3: Two continuous predictors

Now both predictors are continuous. The figure shows the first predictor on the x axis, with point shapes for whether the second predictor is below or above its median. To draw a model, pick a value of the second predictor and draw the line for the first. In the additive model, those lines are parallel. In the interaction model, the slope of the first predictor changes with the second. For each scenario:

1. **Data.** Look at the data in the figure.
2. **Full model.** Write the equation for the interaction model. What does the interaction coefficient mean?
3. **Draw the models.** Use the coefficients to draw each model's line at the two given values of the second predictor.
4. **Degrees of freedom.** Fill in p and df for each model. How many df does the interaction cost?
5. **F test.** Find the F statistic and p-value for the interaction in the `anova()` table.
6. **Interpretation.** State H_0 and H_A. Then interpret the F test in a sentence about the system, not the coefficients.

3A. Algae blooms (together)

Phosphorus from fertilizer and wastewater feeds algae blooms in lakes, and warm water can make blooms worse. A state agency measured **phosphorus** (µg/L), summer water temperature (**temp**, °C), and chlorophyll (**chl**, µg/L), a measure of algae, in 24 lakes. Does the effect of phosphorus on algae depend on temperature?



Full model

$\text{chl} = \beta_0 + \beta_1 \text{phosphorus} + \beta_2 \text{temp} + \beta_3 \text{phosphorus} \times \text{temp} + \varepsilon$

The slope for phosphorus is $\beta_1 + \beta_3 \text{temp}$. β_3 is how much the phosphorus slope changes for each 1 °C of temperature.

Draw the models

Draw each model's line at 12 °C and at 23 °C.

```
coef(lm(chl ~ phosphorus + temp, lakes))
```

```
(Intercept)  phosphorus        temp
-14.3014933   0.5524749   1.2650350
```

```
coef(lm(chl ~ phosphorus * temp, lakes))
```

```
    (Intercept)      phosphorus            temp phosphorus:temp
     2.17640732      0.05801516      0.20420019      0.03096273
```

Additive model (grey): at 12 °C: intercept 0.9, slope 0.55; at 23 °C: intercept 14.8, slope 0.55. Interaction

model (black): at 12 °C: intercept 4.6, slope 0.43; at 23 °C: intercept 6.9, slope 0.77.

Degrees of freedom (n = 24)

Model	p	df	RSS
`chl ~ phosphorus`	2	22	1,075.0
`chl ~ phosphorus + temp`	3	21	312.6
`chl ~ phosphorus * temp`	4	20	235.4

How many df does the interaction cost?

1 df: the interaction adds one coefficient, β_3.

F test

The F statistic and p-value for adding the interaction are in the `phosphorus:temp` row.

```
anova(lm(chl ~ phosphorus * temp, lakes))
```

```
Analysis of Variance Table

Response: chl
                Df  Sum Sq Mean Sq  F value    Pr(>F)
phosphorus       1 2217.32 2217.32 188.3819 1.223e-11 ***
temp             1  762.38  762.38  64.7712 1.063e-07 ***
phosphorus:temp  1   77.18   77.18   6.5568   0.01865 *
Residuals       20  235.41   11.77
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$ (the effect of phosphorus on algae is the same at every temperature)

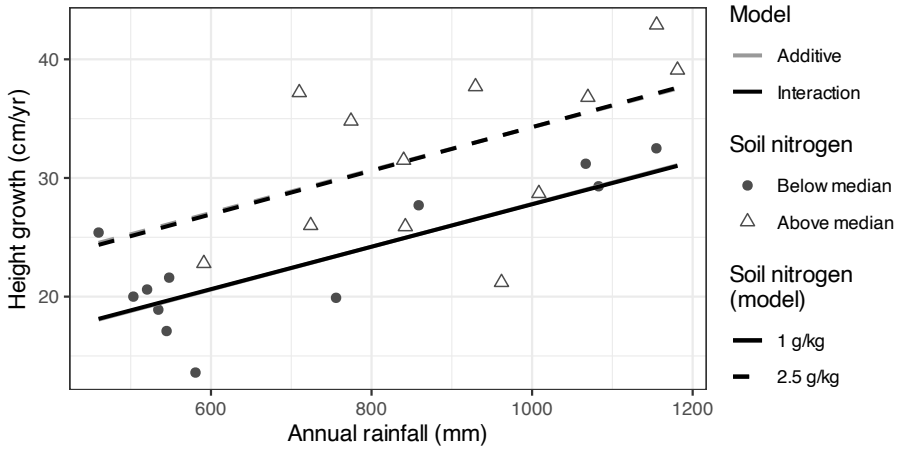
H_A: $\beta_3 \neq 0$

Interpret the F test for the interaction in a sentence.

If the effect of phosphorus didn't depend on temperature, we'd see an F at least this large with probability p = 0.019. We reject H_0. Phosphorus feeds more algae in warmer lakes: each µg/L of phosphorus adds 0.43 µg/L of chlorophyll at 12 °C, but 0.77 µg/L at 23 °C.

3B. Tree growth (in groups)

Trees need both water and nutrients to grow. A forester measured annual height **growth** (cm/yr) of young pines at 24 sites, along with annual rainfall (**rain**, mm) and soil **nitrogen** (g/kg). Does the effect of rainfall on growth depend on soil nitrogen?



Full model

$$\text{growth} = \beta_0 + \beta_1 \text{rain} + \beta_2 \text{nitrogen} + \beta_3 \text{rain} \times \text{nitrogen} + \varepsilon$$

The slope for rainfall is $\beta_1 + \beta_3 \text{nitrogen}$. β_3 is how much the rainfall slope changes for each 1 g/kg of soil nitrogen.

Draw the models

Draw each model's line at 1 g/kg and at 2.5 g/kg of soil nitrogen.

```
coef(lm(growth ~ rain + nitrogen, trees))
```

```
(Intercept)        rain    nitrogen
 5.45629408  0.01804682  4.30840763
```

```
coef(lm(growth ~ rain * nitrogen, trees))
```

```
  (Intercept)          rain      nitrogen rain:nitrogen
 5.9000686445  0.0175474625  4.0022866382  0.0003333131
```

Additive model (grey): at 1 g/kg: intercept 9.8, slope 0.018; at 2.5 g/kg: intercept 16.2, slope 0.018. Interaction model (black): at 1 g/kg: intercept 9.9, slope 0.018; at 2.5 g/kg: intercept 15.9, slope 0.018.

Degrees of freedom (n = 24)

Model	p	df	RSS
`growth ~ rain`	2	22	672.6
`growth ~ rain + nitrogen`	3	21	422.0
`growth ~ rain * nitrogen`	4	20	422.0

How many df does the interaction cost?

1 df: the interaction adds one coefficient, β_3.

F test

The F statistic and p-value for adding the interaction are in the `rain:nitrogen` row.

```
anova(lm(growth ~ rain * nitrogen, trees))
```

```
Analysis of Variance Table

Response: growth
              Df Sum Sq Mean Sq F value    Pr(>F)
rain           1 738.95  738.95 35.0242 8.673e-06 ***
nitrogen       1 250.56  250.56 11.8760  0.002554 **
rain:nitrogen  1   0.08    0.08  0.0039  0.950946
Residuals     20 421.97   21.10
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

State H_0 and H_A.

H_0: $\beta_3 = 0$ (the effect of rainfall on growth is the same at every level of soil nitrogen)

H_A: $\beta_3 \neq 0$

Interpret the F test for the interaction in a sentence.

If the effect of rainfall didn't depend on soil nitrogen, we'd see an F at least this large with probability p = 0.951. We fail to reject H_0. Rainfall and soil nitrogen each speed growth, and the effect of rainfall is about the same whatever the soil nitrogen. The additive model is enough.

Part 4: Design the study

Researchers rarely hand you a predictor and a response. Each client below describes a problem in their own words. For each one:

1. **Variables.** Fill in the table: what are the variables, is each one categorical or continuous, and is each one a predictor or the response?
2. **DAG.** Draw a DAG of the variables.
3. **Models and hypotheses.** Write the equations for the simpler (additive) model and the more complex (interaction) model. State H_0 and H_A for the client's question.
4. **Sketch.** Sketch data and model fits that would support the client's hypothesis. Label your axes.

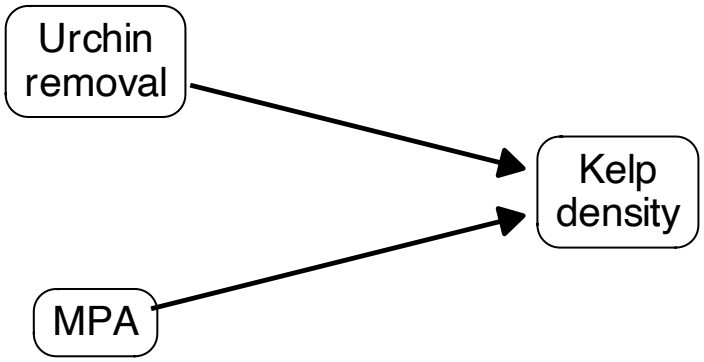
4.1. Urchins and kelp

> "Sea urchins have been mowing down our kelp forests, so for the last three years our divers have been removing urchins from some reefs and leaving others alone. Some of our reefs are inside marine protected areas, where sheephead and lobsters, which eat urchins, are common. Others are outside, where they're fished. We count kelp plants per square meter on every reef. I suspect removal makes a much bigger difference outside the protected areas, since inside them the predators are already keeping urchins in check. Is that right?"

Variables

Variable	Categorical or continuous?	Predictor or response?
Kelp density (plants/m²)	Continuous	Response
Urchin removal (removed or not)	Categorical	Predictor
Protection (inside or outside MPA)	Categorical	Predictor

DAG



Models and hypotheses

Simpler: kelp $= \beta_0 + \beta_1(\text{removal} = \text{Removed}) + \beta_2(\text{MPA} = \text{Outside}) + \varepsilon$

More complex: add $\beta_3(\text{removal} = \text{Removed})(\text{MPA} = \text{Outside})$

H_0: $\beta_3 = 0$ (removal has the same effect inside and outside MPAs)

H_A: $\beta_3 > 0$ (removal helps more outside MPAs). A two-sided $\beta_3 \neq 0$ is also fine.

Sketch



Removal raises kelp a lot outside MPAs and only a little inside, so the difference between removed and not removed is bigger outside.

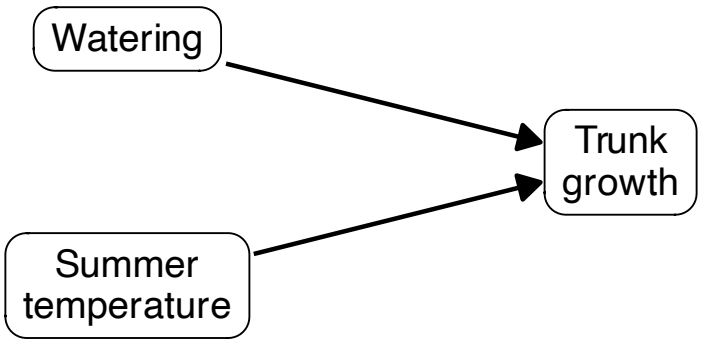
4.2. Street trees

> "The city plants a few thousand street trees a year. In some neighborhoods we pay residents to water new trees through their first summers, and in others we can't afford to. Our crews measure each tree's trunk growth every fall. We also have summer temperatures for every neighborhood from the heat-island study, and some neighborhoods run 8 degrees hotter than others. Council wants to know whether the watering program is worth expanding. My hunch is that watering matters most in the hottest neighborhoods, which would tell us where to spend the money first."

Variables

Variable	Categorical or continuous?	Predictor or response?
Trunk growth (cm/yr)	Continuous	Response
Watering program (yes or no)	Categorical	Predictor
Summer temperature (°C)	Continuous	Predictor

DAG



Models and hypotheses

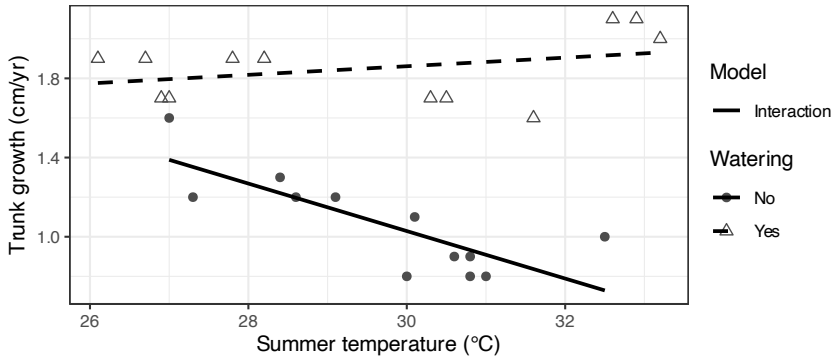
Simpler: growth $= \beta_0 + \beta_1 \text{temp} + \beta_2(\text{watering} = \text{Yes}) + \varepsilon$

More complex: add $\beta_3 \text{temp} \times (\text{watering} = \text{Yes})$

H_0: $\beta_3 = 0$ (watering helps by the same amount at every temperature)

H_A: $\beta_3 > 0$ (watering helps more in hotter neighborhoods). A two-sided $\beta_3 \neq 0$ is also fine.

Sketch



Growth drops in hotter neighborhoods without watering, but stays about the same with watering, so the gap between the lines widens with temperature.

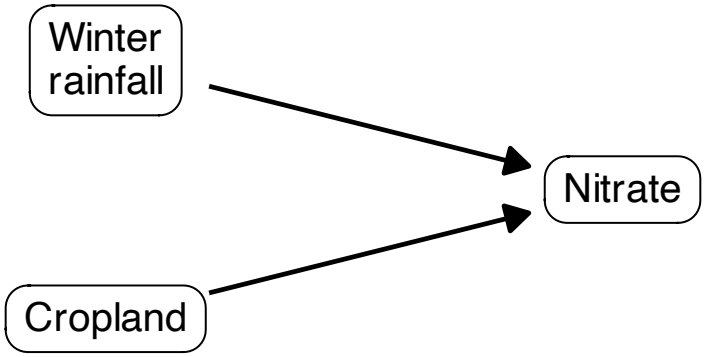
4.3. Nitrate in wells

> "We test nitrate in about forty drinking water wells every spring. Some of them sit in the middle of farm country and some are surrounded mostly by oak woodland, and we've mapped the percent cropland within a kilometer of each well. The numbers jump around a lot from year to year, and it seems like they jump the most after wet winters. My theory is that heavy winter rain flushes fertilizer down into the aquifer, but only where there's cropland to flush it from. If that's right, we'd know which wells to test more often after a big storm year."

Variables

Variable	Categorical or continuous?	Predictor or response?
Nitrate (mg/L)	Continuous	Response
Winter rainfall (mm)	Continuous	Predictor
Cropland within 1 km (%)	Continuous	Predictor

DAG



Models and hypotheses

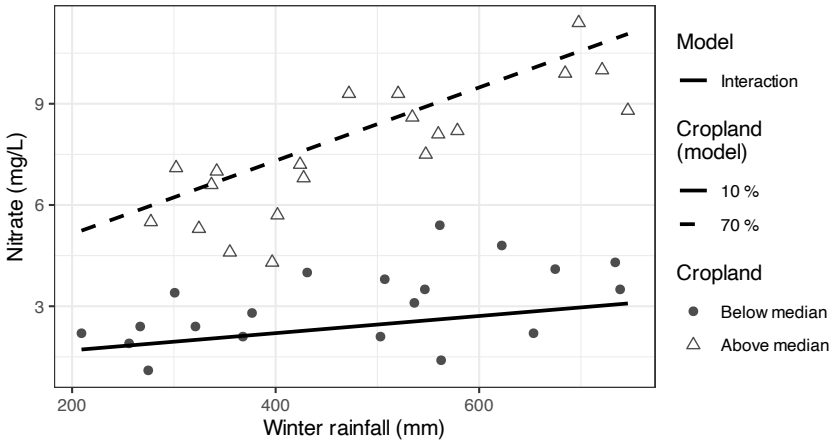
Simpler: nitrate $= \beta_0 + \beta_1$rain $+ \beta_2$cropland $+ \varepsilon$

More complex: add $\beta_3$rain $\times$ cropland

H_0: $\beta_3 = 0$ (rain raises nitrate by the same amount whatever the cropland)

H_A: $\beta_3 > 0$ (rain raises nitrate more where there's more cropland). A two-sided $\beta_3 \neq 0$ is also fine.

Sketch



Nitrate barely changes with rainfall where there's little cropland, but rises steeply with rainfall where there's a lot, so the lines fan out.